Carnegie Mellon University Africa

PROJECT PROPOSAL

Urban Incident Response System

Kavini Nzau(knzau)
Christian Abiyingoma(cabiying)
Nthabiseng Thema(nthema)
Regis Ndahiro Ngoga(rndahiro)

Due: 30 March 2026

0.1 Background

Kigali, the capital of Rwanda, is one of the fastest-growing cities in East Africa, with a census-recorded population of 1,745,555 and an urbanization growth rate of 4% per annum [1]. The city's emergency response is coordinated through the Rwanda National Police (112), the SAMU ambulance service (912) with nearly 600 ambulances [2], and the Fire and Rescue Brigade (111). Although average ambulance response time has improved from 30 to 14 minutes [3], fire brigade responses can take 30 to 60 minutes [4] and patient geolocation remains hampered by inadequate technology [5]. With national road traffic accidents rising from 8,660 in 2022 to 12,189 in 2024 [6] and the "golden hour" principle linking delayed care to reduced trauma survival [7], an intelligent system that maintains a live city map, triages incidents by severity, routes units optimally, and tracks response analytics could significantly improve emergency outcomes.

0.2 Problem Statement

The City of Kigali lacks an efficient system for dispatching emergency response units across its urban road network, leading to delayed incident response times. This project develops an Urban Incident Response System that models Kigali's road network as a dynamic weighted graph to optimize the dispatching of emergency units to incidents across the city. The system must handle up to 10^5 operations and support:

- **Dynamic city map updates:** Adding and removing roads with immediate connectivity tracking.
- **Incident reporting and triage:** Recording incidents with metadata and maintaining a priority queue for dispatch.
- **Optimal unit dispatch:** Verifying reachability and computing shortest weighted paths from unit to incident.
- **Analytical queries:** Counting incidents within time windows and computing average response times.
- **String-based search:** Prefix-based lookup of units and intersections for dispatcher autocomplete.

All data structures will be implemented from scratch in C++ without STL containers, meeting strict latency bounds under a continuous stream of up to 10^5 intermixed commands.

0.3 Solution Approach

The proposed solution is an Urban Incident Response System built around seven custom data structures, each designed to solve a specific aspect of the emergency dispatch problem. No single structure handles everything; each one takes responsibility for the tasks it is best suited for, and they are wired together through a central `Dispatcher` class. Figure 1 shows how data flows through the full system.

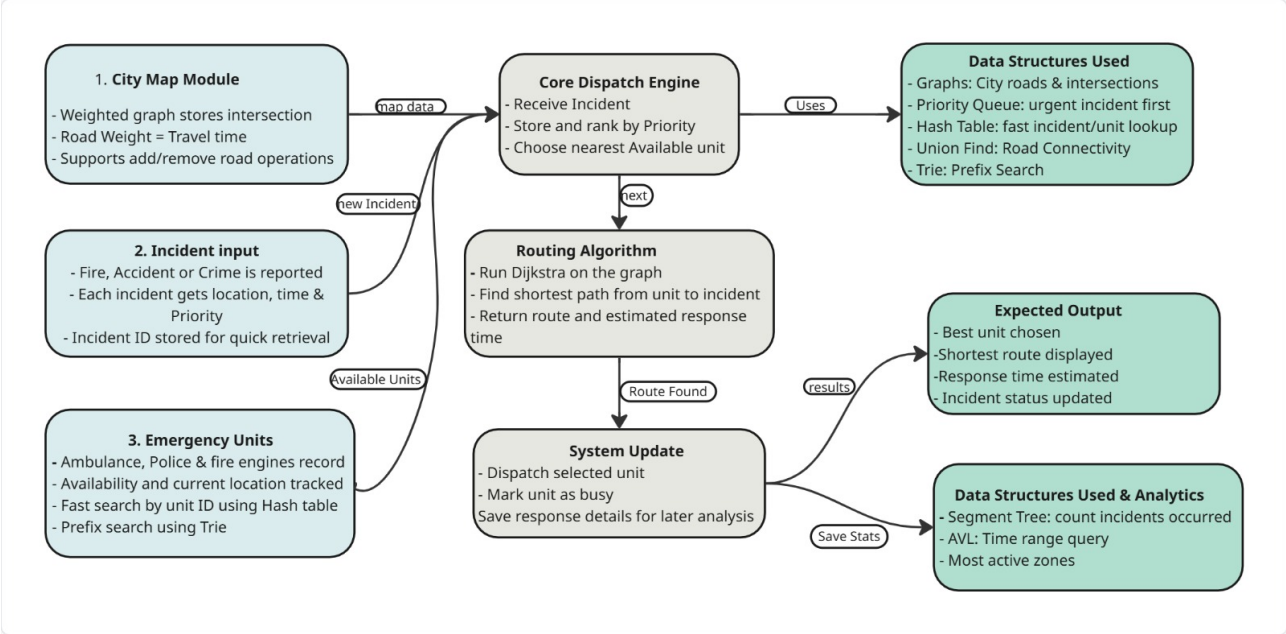

Figure 1: System architecture showing data flow from inputs through the dispatch pipeline, with all seven data structures and their interaction points.

Data Structures

The city map is a **weighted graph** (adjacency list) where intersections are nodes and roads are edges with travel-time weights, supporting dynamic add/remove at runtime. Incoming incidents are ranked by severity in a **min-heap**, keeping the most critical incident at the front; the same heap is reused inside Dijkstra's algorithm. A **hash table** (open addressing, double hashing) stores every incident and unit record by ID for $O(1)$ average lookup. An **AVL tree** indexes incidents by timestamp for efficient range queries. A **segment tree** over 2,048 one-minute buckets answers window-count queries in $O(\log n)$. A **Union-Find** (path compression + union by rank) checks whether a unit can physically reach an incident before routing is attempted. A **trie** stores unit and location names for prefix-based autocomplete in $O(m)$.

Algorithms and Pseudocode

Routing uses **Dijkstra's** shortest-path algorithm. Every dispatch first passes through a Union-Find connectivity check; if the unit and incident are disconnected the dispatch fails immediately.

```
DISPATCH_UNIT(unit_id, incident_id):
  unit     <- hash_table.lookup(unit_id)
  incident <- hash_table.lookup(incident_id)
  if not dsu.connected(unit.loc, incident.loc):
    return FAILED, "no connected route"
  dist, prev <- dijkstra(graph, unit.loc)
  path <- reconstruct_path(prev, incident.loc)
  unit.status <- EN_ROUTE;  incident.status <- DISPATCHED
  return path, dist[incident.loc]
```

Listing 1: DISPATCH_UNIT: connectivity gate followed by shortest-path routing

```
REPORT_INCIDENT(location, type, severity, timestamp):
  id <- next_id++
  hash_table.insert(id, incident)
  min_heap.insert((10-severity)*1e5+timestamp,id)
  avl_tree.insert(timestamp, incident)
  segment_tree.update(timestamp mod BUCKETS, +1)
  return id
```

Listing 2: REPORT_INCIDENT: incident written to four structures simultaneously

0.4 Expected Results

Time Complexity Analysis

Operation	Data Structure	Time Complexity
Add/remove road	Graph + Union-Find	$O(\log n)$
Report incident	Hash + Heap + AVL + SegTree	$O(\log n)$
Dispatch unit	Heap + Hash + DSU + Dijkstra	$O((V+E)\log V)$
Time range query	AVL tree	$O(\log n + k)$
Incident count (window)	Segment tree	$O(\log n)$
Prefix search	Trie	$O(m+k)$
Connectivity check	Union-Find	$\sim O(1)$ amortized
Unit/incident lookup	Hash table	$O(1)$ average

Table 1: Projected time complexities of core system operations

Benchmarking Scenarios

- **Mass incident:** 500 simultaneous incidents across Nyamirambo and Kicukiro, testing priority queue throughput and concurrent Dijkstra routing.
- **Road closure cascade:** Sequential removal of 50 edges along the KN 3 corridor, measuring Union-Find rebuild time and rerouting overhead.
- **Analytics burst:** 10^4 segment tree range queries over a 24-hour incident log, comparing against naive linear scan.

Design Trade-offs

The table below summarizes each design decision and its associated trade-off.

Table 2: Design Trade-offs

Decision	Choice	Rationale	What We Gave Up
Graph representation	Adjacency list	Kigali's road graph is sparse	O(1) edge lookup that an adjacency matrix would have provided
Collision resolution	Open addressing	Better cache performance, simpler memory management	Hash table clustering; linear probing has worse performance under high load than separate chaining
Balanced BST	AVL tree	Stricter balance guarantees faster reads	More complex rotations and slower insertions than Red-Black trees; Red-Black trees would be simpler to code
Edge deletion in DSU	Periodic full rebuild	Simpler than link-cut trees	Alternative dynamic structures support deletions but are more complex; rebuild trades runtime for simplicity
Multiple indexes per incident	4 parallel structures	Speed over memory	Memory footprint is 4× larger; single optimized structure would use less RAM but be slower
Shortest path algorithm	Dijkstra	Simpler implementation, guaranteed correctness without heuristic design	A* would explore fewer nodes and run faster with domain knowledge, but requires a domain-specific heuristic function

The system is expected to handle up to 10^5 operations efficiently while maintaining fast response times, accurate routing, and stable performance under high load. By combining multiple optimized data structures with Dijkstra's algorithm, the solution simulates a realistic urban emergency response system capable of operating under dynamic and high-pressure conditions.

References

[1] City of Kigali, "Overview," Kigali City Government, 2024. [Online]. Available: https://www.kigalicity.gov.rw/about/overview

[2] Rwanda Biomedical Centre, "SAMU Services," RBC, 2024. [Online]. Available: https://www.rbc.gov.rw/who-we-are/our-divisions-and-units/samu-services

[3] Ukwelitimes, "The number of ambulances in Rwanda rises to 510," *Ukwelitimes*, May 14, 2025. [Online]. Available: https://ukwelitimes.com/en/the-number-of-ambulances-in-rwanda-rises-to-510

[4] "Kigali: How fast is firefighter response mechanism?" *The New Times / AllAfrica*, Jul. 17, 2014. [Online]. Available: https://allafrica.com/stories/201407170185.html

[5] M. Hunt, M. Niyonsaba, J. M. Uwitonze, J. D. Nyinawankusi, J. Davies, and R. Rickard, "Challenges locating the scene of emergency: A qualitative study of the EMS system in Rwanda," *Prehospital Emergency Care*, vol. 28, no. 3, pp. 501–505, 2024.

[6] IGIHE, "Over 2,900 people killed in Rwanda road accidents over four years," *IGIHE News*, 2025. [Online]. Available: https://en.igihe.com/news/article/over-2-900-people-killed-in-rwanda-road-accidents-over-four-years

[7] E. B. Lerner and R. M. Moscati, "The golden hour: Scientific fact or medical 'urban legend'?" *Academic Emergency Medicine*, vol. 8, no. 7, pp. 758–760, 2001.